

FUNDAMENTALS OF
**DATABASE
SYSTEMS**

7TH Edition

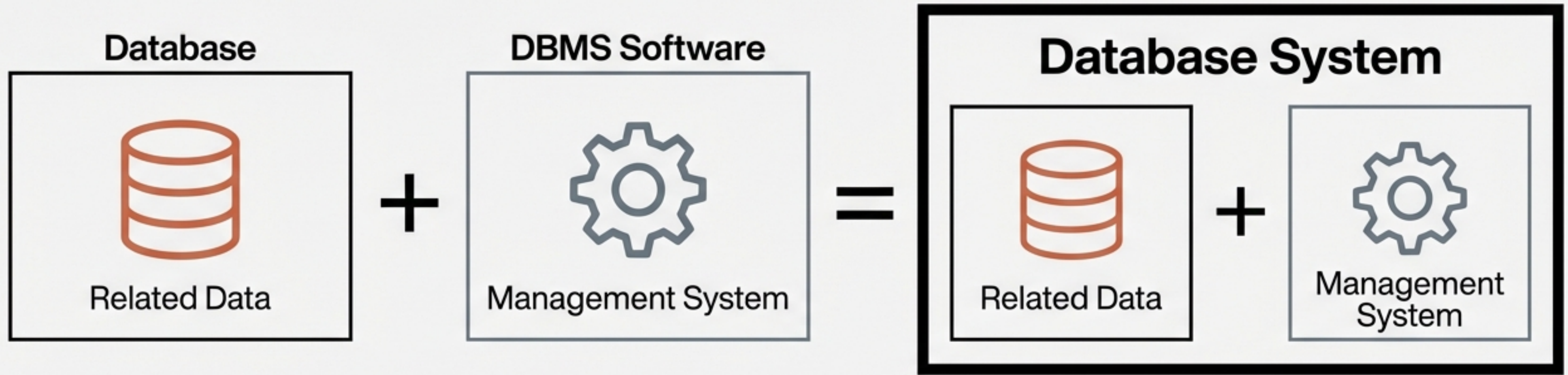
ELMASRI • NAVATHE

Fundamentals of Database Systems

Chapter 01: Databases and Database Users

A comprehensive primer on the architecture, definitions, and advantages of the Database Approach. Adapted from the work of Ramez Elmasri and Shamkant B. Navathe (7th Edition).

READING DECK / SELF-PACED REFERENCE



The Reality (The Mini-world)

The “Universe of Discourse” (UoD). A specific slice of reality—like a university’s student grades—that we wish to capture.

Changes in the Mini-world must be reflected in the system.

The Tools

Data: Known facts with implicit meaning.

Database: A logically coherent collection of data for a specific purpose.

DBMS: General-purpose software to create, maintain, and manipulate the database.

The Anchor Example: A University Database Schema

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to CS	CS1310	4	CS
Data Structures	CS3320	4	CS

Grade Report

The database consists of files (tables) of records, where column headers represent attributes and rows represent entities.

The Landscape: Evolution of Applications

Traditional Applications

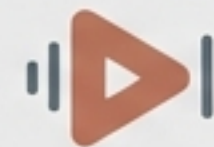
 Numeric & Textual Databases

 University Records

 Bank Transactions

 Airline Reservations

New Wave Applications

 Multimedia DBs (Video/Audio)

 GIS (Geographic Information Systems)

 Data Warehouses & OLAP
(Business Intelligence)

 Real-time & Active DBs
(Industrial Process)

 Big Data / NoSQL
(Social Media, Search)

Main Characteristics of the Database Approach

The four pillars that distinguish DBMS from File Processing

1. Self-Describing Nature



The system stores not just data, but the definition of the data (Meta-data).

2. Insulation & Abstraction



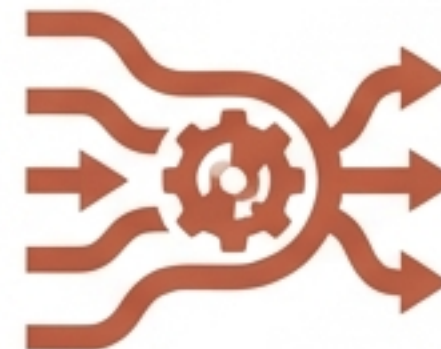
Program-Data Independence.
Changing data structure doesn't break code.

3. Multiple Views



Different users see only the data relevant to their role.

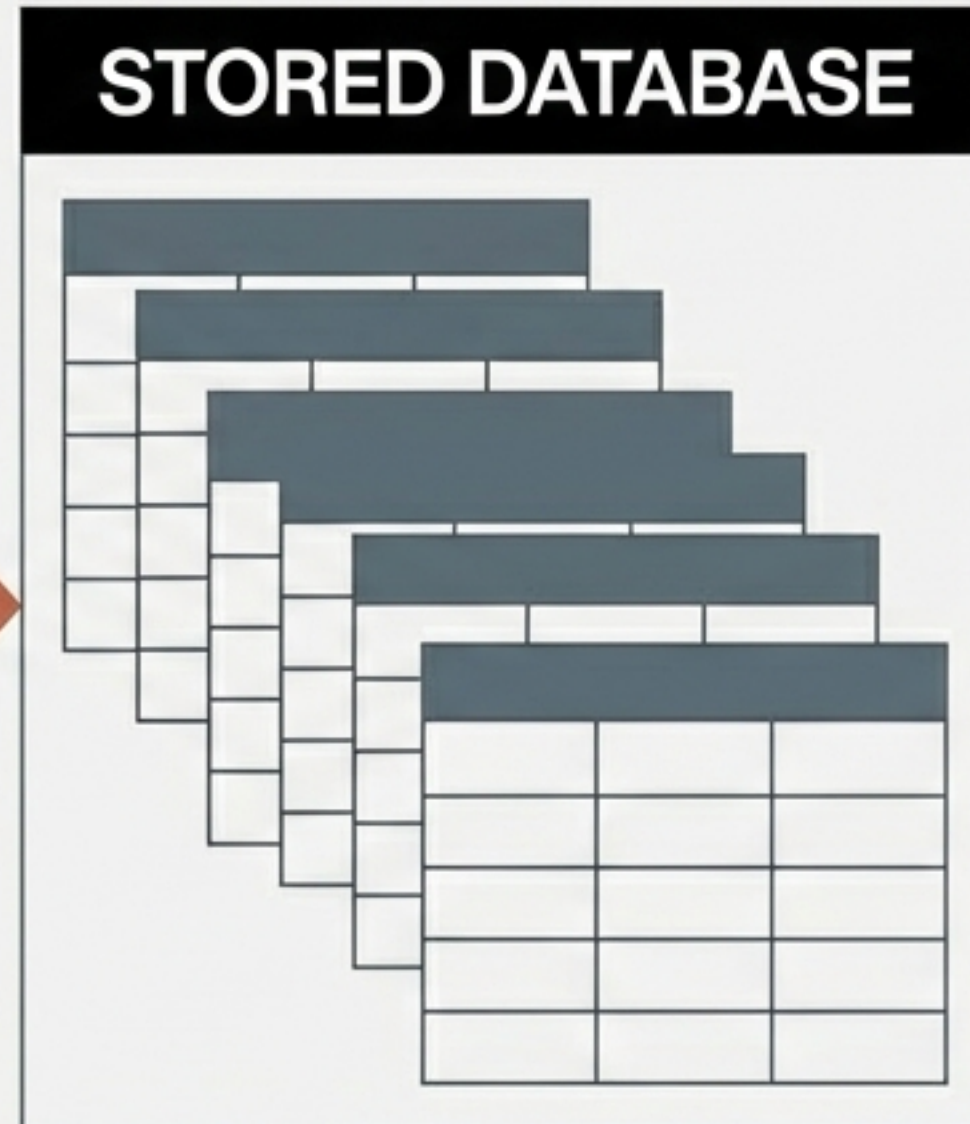
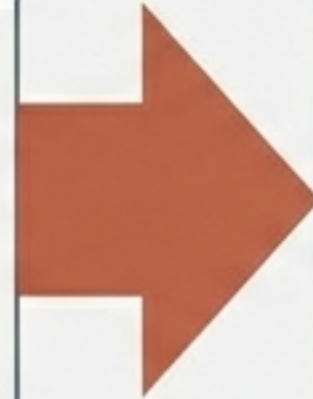
4. Multiuser Processing



Concurrency control allows simultaneous access without errors.

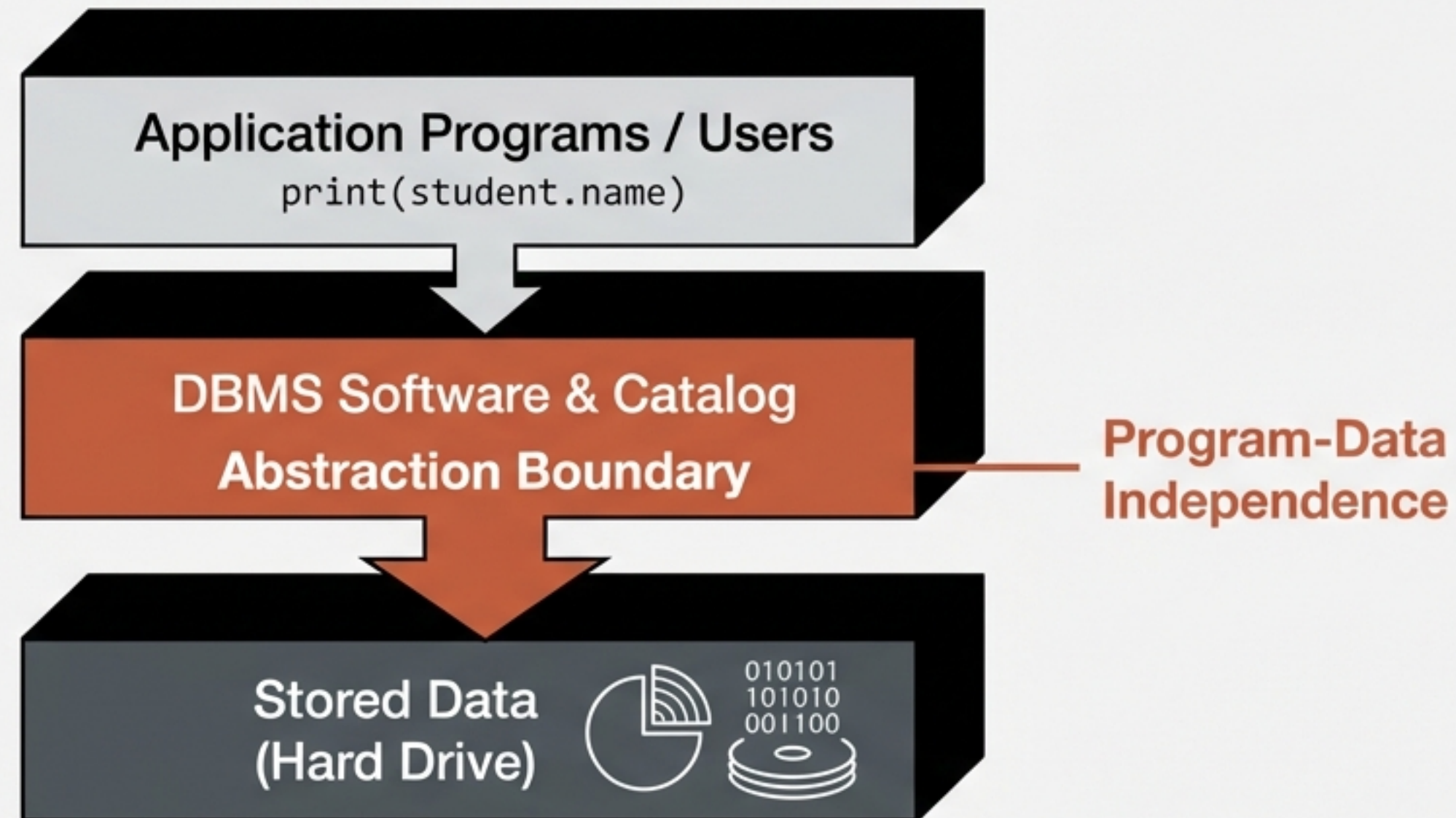
1. Self-Describing Nature: The Catalog

DBMS CATALOG (Meta-data)		
Relation_name	Column_name	Data_type
STUDENT	Name	Character(30)
STUDENT	Class	Integer(1)
COURSE	Course_name	Character(10)



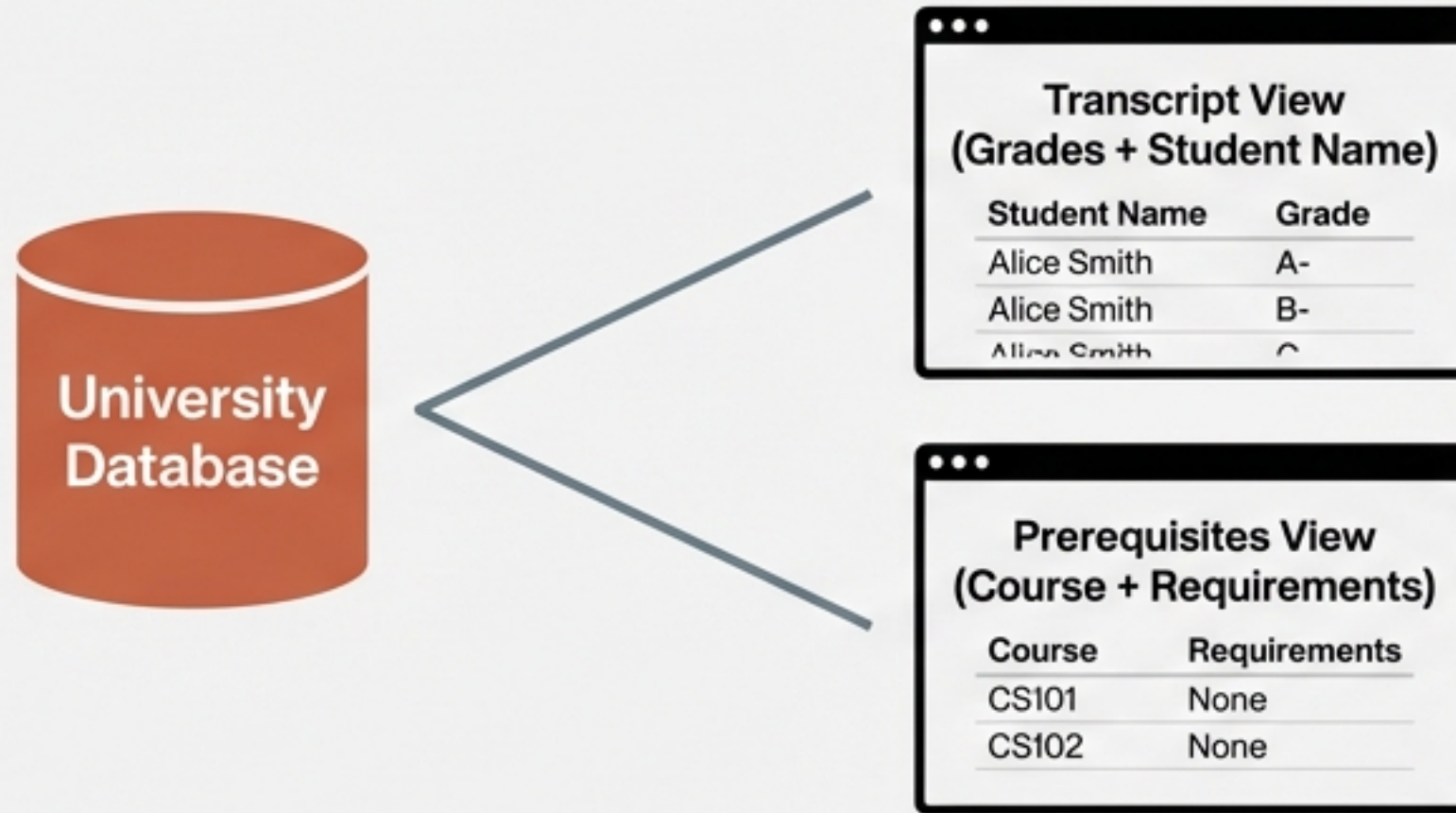
Unlike file processing, a DBMS stores the database definition in a Catalog. This “Meta-data” allows the DBMS software to work with any application (Banking, Insurance, University) by reading the catalog structure dynamically.

2. Insulation & Abstraction



In file processing, code is tightly coupled to file structure. In a DBMS, the code talks to the software, and the software handles the storage. You can change the storage format without rewriting the code.

3. Multiple Views & 4. Concurrency



Users see only derived data relevant to their specific role.



OLTP (Online Transaction Processing) ensures isolation. If two agents book a seat at the same second, the system ensures valid execution (Atomicity).

Database Users: Actors on the Scene

Database Administrator (DBA)

The Chief

Authorizes access, monitors usage, handles security, and acquires resources.

Database Designers

The Architects

Identify data to be stored, choose structures, and communicate with users.

System Analysts & Programmers

The Engineers

Determine requirements and implement "canned transactions" for end users.

End Users

The Consumers

- Casual: Occasional access.
- Naive: Bank tellers, clerks.
- Sophisticated: Scientists/Engineers.

Database Users: Workers Behind the Scene

The builders of the tool, not the users of the data.



DBMS System Designers

Implement the DBMS modules (catalogs, query processing, concurrency control).



Tool Developers

Create design tools, performance monitors, and GUIs.



Operators & Maintenance

Manage the hardware and computer system environment.

Advantages of the Database Approach (I)

Problem

Uncontrolled Redundancy

Every department keeps its own files.
Wasted space. Inconsistent data.

Solution

Normalization

Logical data items are stored in one place.
Controlled redundancy improves query performance.

Security Risks

Open access to sensitive files.

Restricted Access

Authorization subsystems. Students view grades but cannot change them.

Impedance Mismatch

Data structures in code differ from storage.

Persistent Storage

Seamless integration with programming objects.

Advantages of the Database Approach (II)



Enforcing Integrity Constraints

Ensures validity. Data types (integers vs strings), Uniqueness (Student ID must be unique), and Referential Integrity (A grade must belong to a valid student).



Standards & Flexibility

Enforce naming conventions and display formats. Change data structures without scrapping software.



Availability & Backup

Recovery subsystems restore data automatically after hardware or software failure.

A Brief History of Data Models



Legacy Era

Hierarchical &
Network Systems.
Efficient but rigid.
Navigational access.

The Standard

Relational Model
(Codd, 1970). SQL,
Data Independence,
Logical Tables.

The Modern Era

Object-Oriented, Web,
NoSQL & Big Data.
Handling unstructured
data at massive scale.

When NOT to Use a DBMS



High Overhead: High initial investment in hardware, software, and training.

Unnecessary Complexity: Generality slows down simple, specific tasks.

Strict Real-Time Requirements: When overhead latency is unacceptable.

Static Data: If the data is simple, distinct, and never changes.

Chapter 01: Summary Cheatsheet

Definitions

- DB + DBMS = System
- Mini-world = Reality
- Meta-data = Catalog

Key Characteristics

- Self-Describing
- Insulation (Program-Data Independence)
- Multiple Views
- Multi-user Concurrency

Actors

- DBA (Admin)
- Designers (Structure)
- End Users (Casual/Naive/Sophisticated)

Top Advantages

- Redundancy Control
- Integrity Constraints
- Security
- Backup & Recovery